

ActInf ModelStream #002.1 ~ Noor Sajid & Philip Ball

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hello and welcome everyone to the active inference lab this is the model stream number 2.1 on april 16 2021 and today is going to be an awesome model stream we're just going to briefly go around and introduce ourselves and then i'll just mention how the session will be run today and then we'll pass it to noor for a presentation so i'm daniel and i'm a postdoctoral researcher in california i'll pass to philip hi yeah i'm currently a phd student at oxford in my second year and yeah i guess this is work i did kind of before i started my phd um alongside nor and i've currently more focusing on data efficiency within specifically reinforcement learning but i guess on that topic like a knowledge of active inference is obviously useful for approaching such research problems who and or hi hi i'm noah i'm a third year phd student in the theoretical neurobiology group at the wellcome center for human neuroimaging um at ucl um so that's university college london um so my phd supervised by cars focused on these ideas pertaining to adaptation one of which i'll be focusing on today which is behavioral adaptation in non-stationary environments using active inference so thank you awesome thanks both for joining and for this presentation we're going to be hearing a who knows how long presentation from nor and then i'm gonna be compiling questions from the chat so please just type questions as they come to you and then we'll address them at the end so thanks again and nor please take it away perfect thank you um so today i'll be presenting some work um that i did in collaboration with philip who you've just heard from thomas parr and carl fristan um so it's titled active inference uh demystified and compared okay okay perfect okay so the presentation structured as follows can you hear the screens i think it's sharing or was it are you not able to see it i'm not seeing it could you just re-share it yep sure um technology i tell you there we go and i'll crop it so go for it thanks perfect thank you so the presentation structured as well as first i'll briefly motivate the problem setting and provide details of a particular active inference instantiation under consideration today which is the discrete state space setting and the second half of the presentation is going to be focused on some particular examples comparing the active influence formulation with reinforcement learning specifically q learning and bayesian model based algorithm and then what i'm going to do is provide some

face validity of particular aspects of why you would want to even use active inference okay um so what is active inference um it's a first principles account of how biological or artificial agents may operate in dynamic non-stationary settings it stipulates that these agents in order to maintain homeostasis reside in attracting states that minimize their entropy or their surprise so if you take this particular example that we're seeing of this past little hungry agent opening the fridge the way it would work is like you would need to why would you open the fridge right so you want to make a particular choice between eating at home or outside and in order to do that you have to decide what is the optimal action that would allow you to resolve your own uncertainty about the current stage of affairs so um and that would help you then decide whether you want to cook at home or you want to walk to the restaurant and this particular instance this has led to the agent opening the fridge to check whether it even has food at home and what's nice about active inference is that it allows you to think about these problem settings in a more formal way by specifying that optimal behavior is determined by evaluating the evidence that is the sensory input under the agent's generative model of observations that it is being exposed to and in this particular presentation what we'll do is focus on just the process theory that underwrites active inference and not talk through the biological or the neural plausibility of the active inference message passing scheme um but to properly uh motivate why we would want to even use um active inference in comparison to generic reinforcement learning algorithms we need to first start off with this understanding that with an active inference there's this commitment to a pure belief based scheme which means that reward functions are not always necessarily um sorry unnecessary because any policy that you would have has epistemic value even in the absence of preferences additionally active inference agents can also learn their own reward functions and this helps the agent describe the type of behavioral that it expects to see in itself as opposed to something that it would get uh from the environment and these two particular points are really important in contrast to reinforcement learning because under standard rl settings the reward function uh would define how the agent interacts or behaves within a particular environment setting um but defining that reward function in the first place is quite difficult because it assumes that there is a specific signal that's being given from the environment that can be unanimously good or bad for the agent which wouldn't necessarily hold true in a real setting where these um environment signals can change depending on the setting for example eating ice cream is not always going to be rewarding if you're ill and it might make you worse um and that's why constructing these reward functions in the first place is extremely difficult and if you're not constructing them in

appropriate way even with an rl setting it can result in sub-optimal behavior for your agents um an active inference is really good in the sense because we are actually um replacing or bypassing the traditional reward function that you would have in the rl setting with prior beliefs about preferred outcomes so the sort of desired states of affairs that you want to see yourself be in and this becomes important in settings where there's no reward or there's really imprecise understanding of what a reward or preference setting should look like and in this scenario within the standard active influence discrete state formulation what we can do is learn the empirical prior distribution over these preferred outcomes um and intrinsic uh sorry the internal reward function of that agent and this sort of brings me to like the first distinct um conceptualization between rl setting and active inference because within active inference rewards are nothing distinct they're just a standard observation that the agent is getting from the environment whereas in rl they're quite necessary to have appropriate action um that the agent is going to learn um the second point i we want to make sorry we wanted to make was that active influence provides a principal account of epistemic exploration and intrinsic motivation as minimizing uncertainty and again within the rl setting this is quite crucial because the whole uh premise of like lots of new algorithms that we see in our nrl is to try and find the right trade-off for the balance between exploration and exploitation so what are the right set of actions that the agent should make at a given point in time should i carry on choosing all the different ice cream flavors that it's been never been exposed to um like mustard etc or should it always use this oh sorry always have the same ice cream flavor that's been exposed to in the past that it really likes like for example hazelnut or nutella for instance um so this um is it's an outstanding problem with an rl and under a bayesian framework active inference deals with it naturally um using um the expected free energy formulation that i'll come to in a moment um and the last bit that you can see within the active inference framework is that it naturally accounts for uncertainty as part of the belief updating process okay so now that i've sort of um laid out the three things that are super interesting about the active influence scheme in comparison to rl i'm going to provide some intuitions as to why um you can sorry some motivations as to why we can even formulate the active influence formulation the way we do okay so um sorry i just realized i like skipped ahead um no no sorry great presentation yep thank you so much and let me just scroll down to um okay so i previously stated that with an active inference it stipulates that agents are maintaining their homeostasis by residing in attracting states that minimize surprise so you must have been thinking what is surprise well um how we define surprise as a negative log probability of

outcomes and for this we introduce one random variable which is o that corresponds to a particular outcome that's received by an agent and this o exists within a finite set of all possible outcomes and that's o here so o is the first equation that we have just o formally states that $o \in \mathcal{O}$ and here p denotes the probability distribution over outcomes o okay so in active inference the way the agent will actually minimize this surprise quantity that we just walked through is by maintaining a generative model of the world z and this is important because at any given point in time the agent wouldn't necessarily have access to the true measurements of the current state of the world so in this particular graphic that you see here you've got the environment and the agent z interacting with the environment in a particular way it's being exposed to the sensory signal but it doesn't know what the outcome of the o was regenerated by so it can only perceive itself and the world around it through only the o and it needs to make inferences about what type of states or z the true causes z were z responsible for the particular sensory input that it's being exposed to and this is why z in active inference when we formulate the problem we formulate it as a partially observable markov decision process because z in this way we are able to formulate a generative model that it defines this internal distribution over the internal states that the agent would use in order to infer the outcome so he doesn't have access to the true state but it can make hypothesis or beliefs about the states that could have given rise to a particular sensory outcome o that is being exposed to z and using this the agent will make inferences about the true state using a process of reverse mapping specifically bayesian model inversion and to make this a little bit more concrete what you can do is think about the hidden states as locations or color for example and the observation space that the agent would be exposed to would be o for example the velocity of the movement or a particular reward o or like a happy face that they're being exposed to o okay so if we were to think about this a little bit more formally so what is the generative model so as we z described before a generative model is a partially observable mdp within this active inference formulation which rests on a simplified setting that we're considering here where we only have two random variables the first one is o that we've discussed and the second one is z where z denotes z a random variable representing hidden or latent states and they exist within a finite set of all possible hidden states which is denoted by capital z here and this joint probability that we get over o and z can be factorized into the likelihood function which is $p(o|z)$ and then you have the prior over the internal states which is $p(z)$ so this gives you a very nice formulation z that we're going to use in a few next couple of slides and i just wanted to ask are you able to see my mouse when i highlight or

when i or is that not there i can see it yes oh yeah perfect so we know that for an agent to minimize its surprise we would need to marginalize out all the possible hidden states that could have led to a given outcome and this can be achieved by using the factorization that i just mentioned the likelihood and the prior u_m but the the problem is that u_m this is not a trivial task because the dimensionality of the hidden states u_m can be extremely large and if you're considering additional random variables that we're going to introduce in a bit this becomes even more problematic u_m and that's why we use uh another quantity a variational approximation of this quantity uh p of i which is more tractable and allows us to estimate the quantities of interest u_m so so this will u_m be a natural step to talk about the variation free energy which is this variational approximation of the quantity of interest u_m so what is variational free energy so variation free energy is defined as the upper bound on surprise so the first definition sorry first definition that we considered is derived using jensen's inequality and is commonly known as negative evidence lower bound in the variational inference literature so we get this from the equation 4 that we just saw by introducing negative log on both sides and then u_m multiplying this term by 1 which is essentially q of s over q of s so we're assuming that q of s cannot be equal to zero and with that u_m we then apply jensen's inequality and we move the log inside the function and we end up with our expectation with respect to t of s for log of the joint over the approximate or the variational uh quantity of interest here and then we take the negative inside and we can flip it around and we get our first nice quantity of interest here where we get the the bound that we're interested in and in terms of tail between the approximate and the joint that we have so to make this a little bit more concrete what we can do now is to further manipulate the variational free energy summons u_m into uh the kill between the approximate and the true posterior minus the the log evidence that we the model evidence that we had and we can re arrange the last equation to really hone in on the connection between surprise and variational free energy u_m so if you remember that kl is a divergence which means that it cannot be less than zero so u_m it's always strictly uh greater than or equal to zero which means that when our approximate is equal to the true posterior we end up with the variation free energy equal to the model evidence which means that minimizing free energy is essentially equivalent to maximizing the gender model evidence u_m okay u_m we can rewrite the the previous equation that we had u_m equation 10 to express the variational free energy as a function of posterior beliefs in multiple different forms so i'm just going to focus on equation 12 here which is the complexity minus accuracy so this is a trade-off uh that's normally used when the papers that essentially says u_m the complexity tam or

complexity cost is essentially the difference between the approximate entropy of s given π_i with respect to the entropy of s given π_i and here π_i is just your policies and these can be regardless of hypothesis of how the agent is going to act but I'll come back to what policies really entail but for now just consider them as a term that allows us to condition the free energy on a sequence of trajectories of interest and the second term that we have is the log probability of o given s so the likelihood with respect to the key of s that gives you the accuracy so a simple way to think about it is that this is just how accurate the model is and this is some regularization term that's a penalty term to make sure that it's not diverging too far away from our initial priors okay so this particular quantity are there any questions at this point about the variation for energy not yet thank you okay perfect okay so the variation for energy is giving us this way of perceiving the environment and addresses one part of the active inference formulation which is making inferences about the given world that the agent is interacting with at a given point in time however we have not actually accounted for the active part whereas like this particular agent that we have under the active inference formulation can take series of actions or interact with the environment in such a way that it affects that environment in the future so to motivate this a little bit further what we can think about is that not only do we want to minimize our variation free energy we also want to minimize a quantity called expected free energy which depends on anticipated observations about the future in the future or about the future and minimization of this particular terms allows the agent to influence the future by taking particular actions in the present which are selected from a set of policies so I mentioned policies a few times now so what are they so policies can be defined as a sequence of actions at time τ that enable an agent to transition between hidden states and τ here is essentially as a sequence of trajectories up to a particular horizon cap which is considering the total number of time steps that you are considering in a particular setup and for us to properly define a policy we need to introduce two additional random variables so the first one is an action that's conditioned on τ which is denoted by u_τ here and this exists within a finite set of all possible actions that the agents can take and the second random variable that we introduce is a policy which is the π_i that we've discussed all possible policies or sequence of actions in a sense of like the sequential policy optimization that we're interested in here so to make it a little bit more concrete the random variable can be decomposed into a series of actions over a particular time horizon τ so u_1, u_2 and u_m going up to u_τ would denote the action from action at time point one action of time point two and so on and the link is explicit

when you consider that if you consider a policy at a particular time point so τ then the action that you get would be that action okay cool um so i also wanted to highlight that this definition of policy is actually quite different to or distinct to how it's considered in rl which is um when they say policy that means state action policies so as i just mentioned an act of inference a policy is simply a sequence of choices for actions through time that is a sequential policy and this is different to a state action policy in reinforcement learning which is mapping of states to actions so your rl policy which takes in account the action and the state is probability of your action given the state and under um our mdp formulation the definition of action oh sorry definition of policies in rl and active inference become exactly the same when we consider the setting where τ is equal to one so you're only considering one step ahead okay okay so um i'm going to move a little bit and consider uh the quantitative interest they expected free energy which is how do we even derive it so in order to derive it we first need to extend the variational free energy definition that we had before so a few slides ago and now make it dependent on both time so τ and policy and what we're doing essentially is taking that same equation and just decomposing it for previous sorry previous and current time step under a particular policy so that way this is why we have the conditioning and then we're decomposing it in a specific way um in equation 15 and then writing out the matrix formulation in equation 16. so we can come back to this if there are any questions but the key thing to take away from the slide is that now we're including a functional dependency on time um for the variation for energy um and this is allowing us to now move to the expected free energy formulation um but the key thing to note here is that we're only considering time points um the previous time point and the present not the future at all so using the free energy equation before we can derive uh the expected free energy um and what is the expected free energy then so expected free energy is um the free energy function of future trajectories g and it effectively values the evidence for plausible policies based on outcomes that have not been observed yet so that's the key thing so you're making inferences about the polls the set of future trajectories that you haven't observed um and there's a two heuristics that are introduced here in order to get to the g formulation that we see in equation 17. the first one is to include beliefs about future in the expectation that is we're supplementing expectation under the approximate posterior with the likelihood here which results in a predictive distribution given by these first two terms here and the second one is that we're implicitly or i guess explicitly conditioning on the joint probabilities of states and observations in the gender model um sorry in the gender model um dependent on the desired state of affairs as opposed to a particular policy now so this constrains the type of

preferences that the agent would have and what's helpful with these two moves that we're making is that we can now evaluate this quantity before actually having the observations is that the minimization of g would actually encourage policy policies to be consistent with the desired state of affairs that the agent expects itself to be in um i'm just going to briefly mention that this is not the only way to derive expected free energy and there's been some work um that's looked at other formulations uh including work by Carl um where the formulation of the expected free energy can be decomposed into different um structures so if anyone's interested in that we can go through it later um but this free energy expected free energy that i just introduced can be decomposed in certain ways um so equation 20 and 21 give two different decompositions of the first one being the epistemic and the extrinsic value trader and the second one being the expected and the uh cost and ambiguity term so if we just consider the um the first equation we can say that if we're minimizing this equation then we're capturing this imperative to maximize the information gain that you would have from observing the environment um about particular hidden states while maximizing the expected value which is scored by the log preferences or the extrinsic value here so this particular formulation actually gives us a very clear trade-off between the um the first component which is the epistemic value that promotes curious behavior so that's what you want with exploration encouraged as the agent seeks out these new states that minimize uncertainty about the environment and the latter bit is more pragmatic and it encourages exploitative behavior through um this understanding of the type of uh policies that the agent would prefer to reach um in other words like this expected free energy formulation and that we're seeing in equation 20 is essentially treating exploration and exploitation as two different ways of tackling the same so minimizing uncertainty that mentioned at the start of the presentation okay um we can also think about the second equation here which is just offering us an alternative perspective on the free expected frequency which is an agent wishes to minimize the ambiguity and the degree to which outcomes under a particular policy deviate from prior preferences thus ambiguity here is the expectation of the conditional entropy or the uncertainty about outcomes under the current policy in this particular setting low entropy would suggest that outcomes are quite salient and uniquely informative about the hidden states so for example the visual cues that you might see if the room is actually completely tough in comparison to if it's quite dark you're not going to make out anything important from that in addition um the agent will like to pursue policy dependent outcomes that resemble its preferred so it's the donated by c here and this is achieved when the care divergence between the predicted and the preferred outcomes is

minimized by a particular policy okay um and these prior beliefs about the future outcomes equip the agent with goal-directed behavior which is one of the i guess the instances that is really important in active inference um okay so once we have the expected free energy we can derive um the the policies um so um and this is realized by deriving the probability of any policy by applying a soft max function over the expected free energy and this sort of illustrates the self-evidencing behavior of active inference because any sort of policy or action sequence that results in lower expected free energy are more likely and intuitively this would uh make sense because the expected free energy is sort of encapsulating all the types of things that you want to include or consider when you're interacting with the world so you want to explore you want to exploit but you want to have a balance of that um and then when you're selecting your policy um it's just a matter of determining the actions which get you closest to this particular core um and this can be defined by an attractive state that's defined by your c matrix that we described before um and if you don't have that then it's just random exploration that you would get um sometimes you can also include a temperature parameter β here um and by having a hyper prime on this you introduce an additional complexity cost into the uh formulation which allows you to account for um how flat or how confident or precise you want your preferences to be over the policy space um the key thing to note is that um for a sake of simplicity i'm not going to go through a lot of the the details about how these are optimized but you can do that in multiple different ways for example in active we can optimize expectation about the hidden states of interest the policies the precisions that are inferred and then we can also optimize the model parameters through um through the learning procedures involved um but those sort of differ depending on the setup you're looking at for example if you're using variation base you would just iterate this uh these functions or objective functions until convergence or an active inference um you do a gradient ascend um to find the sufficient statistics of interest again this depends on exactly which formulation and setup you're looking at but the key thing to note here is that um that there's three particular aspects to the active inference algorithm that are useful and can be uh taken from this particular uh framework and applied to uh other settings um so i'm just gonna reiterate and summarize them briefly so we first have the generative model that's crucial so in order for an agent to interact and minimize its surprise it needs a generative model of the world and that's described as simply i'm not i'm not including any of the model parameters here but you can have your outcomes your states and your policies and this um these are decomposed into um yes sorry there's a arrow with the brackets here but these are decomposed into your prior your likelihood and your transition and then you set once you

have this genital model the objective the agent is to fit the model to sample observations to reduce the price and that is through variational free energy optimization so this particular trade-off that we have between the complexity and the accuracy cost and then the second oh sorry the last part of this algorithm is to plan so select actions that minimize uncertainty that um that is the expected free energy and the way you do that is by having a soft max over that this uh g negative of g the quantity we have here and then sampling from that in order to select the next specs best action okay so that's a quick and deep dive into a massive amount of active inference literature but i just wanted to highlight that those are the three core ingredients that if you are interested in implementing these algorithms yourself okay so now i'm just going to switch gears a little bit and walk through comparisons with reinforcement learning so in our work we considered a modified version of the open ai gyms frozen lake environment so frozen lake has a grid-like structure with four distinct patches so it has a starting point which is s um so we can see it here um apologies but it's super tiny so s is here and you've got the frozen surface which is f so again i don't think i can differentiate because i'm just moving my mouth um i'm zooming in they can see it perfect so you've got the frozen f and then you've got the whole and then lastly you have the goal so g here where the first b is located and all patches in this particular setup are safe except for hall where if the agent goes to h it gets a negative reward um the agent starts each episode at the first um position which is the starting position and from there it needs to reach the frisbee location in um in the least amount of steps possible and the way you can do that is by performing uh four different types of actions so either going left right down or up and the agent is allowed to carry on moving through the frozen lake uh with multiple revisits so you can go back to the starting position having like on in other places but each episode will end when either reaches the hall or the goal location and these locations differ depending on the setup that we'll um have in our simulations so in one set up um the position of the whole is eight and the goal is six and another set up the position of the whole is six and um the goal is eight and the objective is as i said to reach the goal in ideally as few steps as possible while avoiding the whole because that would end the episode uh if it reaches the goal it gets a positive reward of 100 and negative otherwise um the key thing to note here is that this query metric actually allows us a way to compare the active inference algorithms to reinforcement learning algorithms but it's not really important for the active inference algorithm to um have the reward function as a get-go because it can still move around and using just the information game tab so not having the extrinsic value component um and that that's quite interesting because we'll see the ramifications of that in our assimilations and for

this particular setup we limited the maximum number of time steps for each episode to 15.

okay so what i'm going to do is first talk through the genital model that we use for the active inference formulation um so here what you're seeing on the slide is a graphical representation of the active and for instance model so this model contains four action states so right down up and left and these control the ability to transition uh location fact so for example if you are in position one and you take the action right then you'll end up in position two or if you're in location five and you take um the up action you'll end up in position two as well um in this particular setting both positions six and eight are absorbing states because if you'll remember once the agent goes to that location they're not able to move out so that's when the episode ends um and if an agent makes an improbable move in this particular maze for example if it tries to go from position one to left it will just stay in that location it won't move in this particular um gender model so i'm just looking at the hidden states now we have a chronic um transfer product between the two factors so we have location and context here um the context cannot be changed by the that we have um because this is something that's determined by the environment and this determines where the goal of the whole locations are whereas the location is something that the agent has control over and that's where we have the action states over this um so with the context we have two contexts the first one is where the goal is in your location eight and the hallways in location six and the second context is where the goal is in location six and the hall is in location eight um at each time point the agent will observe uh two outcomes one would be its own position in this particular maze and the second would be the score that the agent would get um the likelihood for the grid position is entirely determined by the location of the agent and the score um is determined by both the location and the context in place so if the agent is in location six and it's in context two then it will receive a positive reward otherwise it will receive a negative or mutual reward depending on where it is um based on um trying to make that comparison with reinforcement learning what we're doing here is we are introducing uh prime preferences where the agent has plus four um for um positive reward negative four four sorry minus four for negative reward and otherwise and it at the first stage it expects itself to be in the first location so we compared um this particular gender model and the active inference agent to two reinforcement learning algorithms so the first one was the q learning using epsilon greedy exploration and the second one was the bayesian model based reinforcement learning algorithm using standard thompson sampling and thompson sampling is appropriate procedure here because it entails the optimization of dual objectives reward maximization and information gain and this is achieved by having this

distribution over a particular function that we prioritize by a prior just by having a prior distribution over it that we sample from okay um so for the two q learning algorithms we um have two epsilon greedy uh parameters so one where it's fixed exploration set to 0.1 and then another one where we have decaying exploration that starts from one and decays down to zero so first we assessed um this um how the agents interacted in a stationary setting where the reward wasn't changing uh such that the goal location was always at six and the whole location was always that age and then we evaluate the performance of the agents um the key thing to take away from here is that both the bayesian rl and the active inference agents are able to quickly learn um where the reward location is and just maximize it out and this is um this is performance is consistent denoted by the really tight confidence bounds that we see in comparison the q learning agents are um so for the one where we have uh fixed exploration it's fairly um fairly good and is able to land where the reward is located but there is um some deviation denoted by that ten percent of selecting a random action um whereas the q learning where we have epsilon is equal to one to gain to zero um the performance isn't the greatest and for the null model the active inference where there is no reward here the agent randomly goes to the hall and randomly goes to the goal 50 of the time but the key thing to note is that apart from the non-model all the models are doing fairly well and seem to be performing okay within the stationary so the next thing no could i just make a point about these experiments as well just because they also do veer away slightly from like the traditional rl like which is yeah great so basically like usually in rl what happens is you kind of have this ambiguity between like training time and test time performance and you usually especially in something like q learning like you just hit the max over the q function or you have a policy that tries to do like some search over the q function and takes that max like given a state um but there's kind of an artificial distinction which is like obviously as you're acquiring data you're making mistakes in the environment you're interacting with the real environment so in order to kind of make that fair comparison between here and active inference where you don't really have this distinction between like training time and test time it's all just interaction the that's the reason why the q learning agent especially when epsilon is fixed to say 0.1 never achieves the optimal policy simply because we're also with 0.1 probability taking a random action so we're not making this distinction like you sometimes would with normal rl train and test time we're all saying train and test some are basically the same thing so however you're choosing to interact with the world is how you should be assessed so that's that's why initially looking at these graphs you might be like hang on like why isn't q learning solving this but um yeah that's just to clear

something up if you're more familiar with say some of the like dpml uh experimental procedure perfect thank you okay um and then um just following on from that so we changed the environment a little bit to make it a little bit more difficult to see whether um the bayesian and the active inference agents might struggle when we start having the relocation change after every few episodes so specifically we swapped the goal and the whole location at um at time point 21 at time 121 141 251 and 451 so you can see that in these figures where the line the gray lines are shown so these points they're all location flipped so like the stationary setting for the first 20 trials all the agents seem to be doing as you would expect so both the bayesian rl and the active inference agents are fairly okay uh the q learning with um the fixed exploration sector 0.1 is doing fairly okay as we saw before and the active learning with the decaying exploration is doing as it was doing before but when you flip it around um such that the goal locations change um what you notice is that with bayesian rl agent the amount of reward or the score it gets is quite low and then we see a phase it then transitions and becomes um to the optimal policy again in comparison to the active inference agent where it instantly after the first trial of doing the incorrect is able to switch over to the active and sorry the appropriate policy and the reason for that is for these uh rl settings where we're considering it as a learning problem you need to first do reverse and learning of where the reward location is and then we learn the new reward location so you're seeing that for q learning uh for both the different epsilon greedy parameterizations and also for the bayesian rl and we're seeing that consistently whereas for the active inference agent because we're treating it as a planning as an inference problem where the posteriors from the previous state are moved over to the prize and moved over as priors and the agent is able to instantly um realize that the current policy that was following the this time step is an appropriateness which is its policy to the next um to the the other one again um the null model as expected doesn't really do much because it's just exploring doesn't really care where the reward or the whole location are okay um phil do you want to add something to this or no i think the same point applies to this one yep definitely okay um so just to wrap up with these two comparisons and i with the non-welder stationary setting um all types of agents um including the bayesian rl and the q-learning would be reasonable frameworks to use whereas with the uh with a non-stochastic setting having an active inference agent might be appropriate way of handling changing dynamics um but the key caveat with that is that you can introduce a lot more additional complexity into the bayesian rl or the q learning or the rl framework in general to allow uh for a way to handle uncertainty but it wouldn't be a natural way of adding it you

would have to sort of uh augment the function or the algorithm in particular ways to justify it um okay so so once we have done this comparison we were interested in is um why would you even want to use active inference um when um you don't have an understanding about the world or you don't have sort of a preference over the type of things that could be done because as we saw the active inference model the null model is just exploring it's not really doing much um and that's brings us to one of the initial points that i introduced at the start of the presentation which is that within an active influence framework we don't really care about having a reward function we can learn we can learn that based on some interaction with the environment um so for that um what we did was we carried out a few different simulations to see how the active inference agent can select different types of policies in the absence of prior preferences and for this we we did three different experiments um where we allowed either or the and all the outcome preferences to be learned over time and saw how the agent interacted when this learning over the different um preferences uh took place and the the way we did it was using uh the conjugacy models where uh because these are discrete state models we have categorical distributions over our model parameters and we're introducing directional distributions on top as a hybrid prior and learning those hyper prior um we can go into the details of this if anyone has any questions but just for simplicity um just assume that all the dirichlet distributions uh were set to flat completely flat and the agents were all given an opportunity to land based on different interactions they had so the first set of experiments like the simulations that we ran were to understand how um the agent would reduce uncertainty about the environment that is interacting with so if it doesn't know the the the the frozen lake how would it interact or explore that lake um and what we see is that um in this particular instance the agent um was just interested in exploring it didn't particularly care about the where the goal or the whole location was um and this is highlighted in a series of different exploration um trajectories that we see so the first one was where the agent falls in the hall um and then another one where it goes around and ends up at the second in the second episode at the goal location and in somewhere it just ends the episode by just going back and forth so this is just floor exploration nothing more to hear um the next um set of analysis or simulations that we ran was to see what would happen if the agent knew about the world but had no preferences of the type of outcomes they expected itself to be in and for this um we ran multiple different simulations and saw that in the absence of um any sort of preferences um the whole could actually become really attractive if it's encountered first so we see that in the first figure where the agent learns to prefer to hide in halls and in the second type of trial that

we saw was where the agent uh exhibited the preference to actually go to the goal location so this is entirely dependent on the instantiation or the type of stimulus that the agent is exposed to initially that determines the type of preferences it would learn to have okay and then the last set of simulations that we ran was just to check what would happen when we interacted with the epistemic imperatives actually resolve uncertainty about the environment that the agent was interacting with specifically the likelihood mapping between the outcomes given the states and the uncertainty about the desired state of affairs that the agent expected herself to be in and what we saw was that um if we allowed um a sufficient number of trials to pass the agent in this particular setting learn to prefer to ride sorry hiding holes after x number of episodes but it had a very distinct um preference over the type of outcomes it expected itself to be in depending on the particular uh a sorry particular time point um so that brings me to the last simulation so i'm just going to wrap up which is um in active inference i started the presentation was saying it's a particular algorithm that gives us very nice um things to consider when we're operating in a bayesian or belief based setting which is firstly the principal account of epistemic exploration and intrinsic motivation that we get from the particular expected free energy decomposition that we went through the second thing that i wanted to highlight was that um under an active inference setting we don't have to explicitly specify a reward function which we saw in the last uh the second set of simulations where the agent can also learn its own reward and prefer to become something that is quite counter-intuitive from an rl setting where the signal from the environment is saying something is bad but the agent's internal motivations of preference allow it to then do something that's quite uh at odds with what the environment is expecting it to do and lastly because of this belief um invasion setting uncertainty is a natural part of the belief updating um so within the stationary settings active inference agents perform as well as reinforcement learning agents however in non-stationary settings they outperformed due to their ability to carry out this planning as inference and that's i think that holds um this is not a conclusive statement but it's a nice way to start thinking about how um if you're scaling up active inference agents to interact on the same type of environments as reinforcement learning agents that might be a little bit hard to resolve because there is some non-stationarity or some weird fluctuations happening in the environment and active inference agents could potentially perform quite nicely if we have this planning as inference set up um and that brings me to the end of the presentation so i just want to thank everyone who was involved with this work called tom and phil and everyone who's helped me think through these interesting ideas that i presented so and everyone also for listening thank

you thank you awesome talk so you can maybe unshare and we can ask a few questions and if anyone who's watching live would like to ask questions they're more than welcome so maybe i'll start with just um a general point that it was awesome to see how clearly you differentiated the reinforcement learning and the active inference paradigm and just one question i had was you mentioned that in the case of a time step of one a time horizon of one there was like an equivalence between the reinforcement learning approach and the active inference approach now people are using reinforcement learning to do planning for the future amidst so how are they accomplishing those kinds of planning and where are some situations where might be able to step into those um settings and potentially do better i think that's already a question a good question and feel free for you to jump in if i miss something but um so within the url setting from what i know if they are considering temporal horizons greater than one then they have hierarchical rl or they have options where they're considering trajectories that are greater than one that allow them to consider a whole sequence of play um before they like um that's rolled out if they select a particular policy instead of like that actually one single action to state mapping um but i i'm not i haven't really worked with options that much maybe phil do you know um yeah so options are like one way to achieve this kind of multi-step sort of um like i guess oversized almost like this inductive bias right that you will like your choices will be kind of contiguous blocks of like more than one step whereas a kind of policy kind of it's just like i'm going to take one step and then i'll have this kind of next state i exist in um but i think what's important to disambiguate is actually you can do sort of planning the model style things because you can view active inference as a sort of model based like in the kind of you know rl paradigm and indeed like within model based like you can have um like planning methods similar to what is done in active inference however i guess like the key distinction here is when you do planning in model based rl you're simply just trying to take you tend to take some sort of suit like kind of mini evolutionary method usually something like the cross entropy method and what you're doing is this kind of search over all actions that you take at a certain point in time that maximizes your kind of reward over say a horizon of say 20 steps um and you can do some clever stuff you can terminate with a q value function um but fundamentally like it doesn't take you away from this sort of definitional difference which is in active inference you're doing this sort of planning over actions even though they might seem superficially similar like kind of basically like drawing lots of actions and seeing which ones maximize some sort of utility in the case of active inference all the useful kind of information and the desired rata about um like exploration and it's and um exploitation are wrapped up in

that maximization whereas in rl it's kind of it's a bit harder to understand um like in the classical sense we're just trying to maximize reward but um you can have heuristics where you say oh but maybe i want to also maximize some notion of model uncertainty and you know it kind of gets a bit um more difficult to naturally integrate all these approaches into the same thing um because like in the example of active influence because everything's a distribution you just put a hyper prior over if you want to you just integrate it out if you don't want to deal with kind of tuning it so yeah that's kind of where i feel like there is this kind of difference cool and what what kinds of settings do you think that those types of of uh action as inference and planning as inference could be utilized like what kind of data sets or questions or contexts are people currently using another type of method but then you're excited to see active inference play a role i think mostly open-ended problems where you don't really have a reward function i think because the way the rl agent is learning how to interact with the environment is through a reward function um so anything where you don't really have that or you have um an environment that that is changing um but i know that there's like a whole host of uh people within the rl community working on like intrinsic motivation or internal motivation so those sort of things do overlap with active influence formulation um but particular paradigms i think for me the more interesting aspect of active influence comes from the fact when you start thinking about um biological agents and if you're modeling a patient or someone with schizophrenia you can with this bayesian framework you can change um the priors to try and see how the person is interacting where i guess it's it could be that their policy like the way they're valuing their policies are broken or it just could be other parts that are different but i think within the rl setting if you are stepping outside the um the standard the game mode right so like magic or gym environments and start going into ones which are more open-ended and you don't have any rewards then active influence could be potentially useful but i i'm a little bit hesitant to say whether it will be better because if you start augmenting bayesian rl with all sorts of interesting components then to a certain extent it will be active in front scaled up um which which is potentially a contentious point um but i i think it it depends on exactly what you were incorporating and that's why for this particular presentation and our work we were quite careful in defining what reinforcement learning meant which is that you have to have this reward function in play and you want to maximize this reward function and anything you include into the rl framework has to have that uh that like the object that objective has to be there but if you if you sort of bypass them say okay i'm just going to add in all sorts of to make the algorithm or the way um the interacts with the environment um i i guess as similar to a

active inference framework or even perhaps even better then you don't that fine distinction between where rl is better or where active influence is better sort of it's not really there for me um because i think both communities from my perspective are working from similar things which is sequential decision making um and with our work is mostly with uh sequential decision making in the face of uncertainty whereas some of our work my rl work might not be focused on that so i think it's when you start drawing the boundaries it becomes a little bit hazy where like things are separate or not and i think i went up on a little tangent um but phil do you want to add anything to that in terms of the environments and the paradigms that useful yeah i think if one of your aims is say you're given an environment that you just have no prior knowledge about how you ought to behave you could argue that you could deploy an rl agent that uses some sort of curiosity or epistemic uncertainty reduction mechanism but i mean i know i know there is like a tiny bit of work about learning priors over reward functions and learning those i'm not hugely like aware but um i think what's important to understand is in the limit of like exploring the entire environment your epistemic uncertainty is going to go to zero right like you um you will have observed everything and then it's unclear what your rl agent is going to do at that point especially you say you have a deep neural network that parameterizes um how you take actions whereas in the case and i think what this paper was really interesting for me to see like when we ran these experiments was actually you just put a prior over your prior preferences and eventually you learn a mode of behavior it may not be optimal but your agent eventually learns to adopt a behavior that is self-fulfilling because it reduces epidemic uncertainty and then all that's left is says well i think these are kind of useful behaviors or at least these are behaviors for me to do in the world and eventually you get quite a repetitive behavior and it might be a more kind of accurate simulation of in the absence of any information like how something intelligent might actually behave in a world whereas in the rl paradigm it's less clear you know once you reduce all that epistemic and certainly about where the value is and you haven't found any like what is your agent really doing at that point but i think my problem at the moment with like the acting from formulation is that this works on like a discrete state formulation right we saw really nice results in that setting but i think if you scale it up active inference agents are going to have the same issues if you're using like amortized inference or something to actually uh approximate your likelihood transition functions which means that you might not get these nice properties that we're seeing at the this small scale um so scaling it up is like a it's a interesting and open-ended problem at the moment scaling up in the right way that you can include like these conjugating models that we

have in um the discrete state formulation and that's how we're like do the last set of simulations where we introduce the conjugate priors to do the learning over the the prior preferences but also the likelihood um so it becomes a little bit hazy how to learn like i have a hyper prior over an entire neural network if you're scaling it up that way um so they must like a little work needs to be done that area and in order to actually show that it's appropriate um that you can have these interesting components then you need to start thinking about how you would include these hyper priors uh because it is not reasonable to say that you're gonna have a hybrid prior over the the parameter space such like the the beta hyper so that hyper prior over the gamma that that would be sufficient in those settings um hyper prior for the way the agent is selecting his actions it has to be over the model parameters and if scaling up means that you're losing that nice way to disentangle the particular model parameters then it becomes very uncertain i i don't know it's it's an open-ended problem for me cool yeah i think high-dimensional problems still represent something that um is relatively difficult especially just due to the kind of you know you know the further we stray from bays like um the less principle it becomes and um you know it's like it's quite a fine line very interesting about how whether within the rl or active inference paradigm there's sort of the sparse skeleton the bare bones at the core and then sometimes these other layers or tweaks are needed and pretty interesting to learn about and also what you had said earlier about how the challenge is planning for sequential action when you're in feedback whether just by moving around an environment so your local environment changes or you're playing a game with a board game is going to change or you're trading on a market sequential action you can't just plan steps one through 100 without at least thinking about some what-ifs and then only having access to limited observational data and planning amidst fundamental uncertainty so i think a lot of those points of contact with the motivations of reinforcement learning and machine learning will maybe bring some more light into active inference and um push some of those frontiers you just mentioned so i have a question and anyone else in the chat can ask a question too um let's say somebody is wanting to learn this and they're actually in a lucky beginner's mind perspective because they might not have been enticed by learning reinforcement learning but they've gotten curious in active inference and excited by your presentation and so what what kinds of computer languages or skills might they want to learn or what kinds of approaches or mindsets would be helpful if somebody let's say weren't coming from a classical machine learning perspective and learning active inference but rather kind of upskilling into active inference what would you recommend either of you to them bill do you want to go first with

that yeah i mean i think it's an interesting way to view a kind of this kind of potential person because i kind of felt like i was somewhat like this person like way back when like nora and i was starting to have these conversations about active and i like the the this paper started off basically as a tutorial i was writing having spent two three months kind of in the evenings reading about trying to like sift through the act of inference literature and i think um you know i think it's fair to say that at times it's unapologetically dense and quite difficult to read so you know without trying to you know self-endorse but i do think reading this manuscript in particular like the whole aim philosophy when we like when we started writing this was to really understand like what is what is happening here like what is this kind of expected free energy quantity like why do we care about it so i know from a kind of theoretical perspective i think this is a very lucid presentation of the concept so at least you can get some sort of intuition as to what's happening um as for the kind of coding side i can't really speak to that i'm sure i'm sure nor kind of has has worked it works with it quite a bit um i i was going to say it sort of depends on what the primary objective of the person is is it for a sort of get an understanding of the the high-level conceptual ideas or treat it as an algorithm because if you're going coming from a free energy principle too active inference is a different story or if you if you're taking active inferences of siloed a specific kind of sequential decision-making uh scheme right um so depending on that is sort of differs uh but as phil was saying i think this paper definitely is really nice in the sense like it does try and define all the different concepts and goes through the different formulations maybe not in as much detail um with the assumptions in play it gives you the the layout of how you might be able to derive it um but certain things like what what the approximate density even really entails those are very uh difficult questions where you would have to like dive into variational inference literature to understand so i guess from that perspective someone who's coming into the field should spend some time thinking about variational inference and how that really ties back to the act of inference because the perception part of active inference is in most instances exactly the same as variational uh inference literature and optimizing uh the model evidence uh or the maximizing the evidence law about um this is the second thing that i was going to add is um that the paper with lance de costa is really good for someone who wants to drill down into deriving everything themselves uh sometimes it's super technical so art that the paper that i walked through today uh the one with phil and tom and carl is it's a nice introduction for someone who's not familiar with the mathematics and just wants to get a layman's summary uh whereas this the paper the the paper where lance's first author gives like the detailed

derivations and a lot of the assumptions in place so that's from like understanding the theory part um from the coding perspective it entirely depends what the end objective is so if they want someone wants to work with discrete state formulations then the matlab code carl has written and it's years of work with lots of um nice simulations and examples that you can use and also our code is online and you can access it so there's a link in the paper um in the software um section that gives where exactly the code is and you can look through that and see how the simulations were done if someone is interested in um more i guess um high-dimensional formulations of active inference that um there's some recent work with uh zephyr zephyrus um so zeff fontes as first author uh where we've got a nice again we have a git reaper for that work as well which uh gives uh a breakdown of how you would implement a simple active inference agent using uh very short encoders and a simple transition network so there's lots of like different areas but for someone starting out they sort of have to understand whether they want to focus on the theoretical side or whether they want to focus on the implementation side theoretical side would be drilling down into the variational influence and the maths behind it and if they want to focus on the coding aspect then they want to figure out whether it's continuous or discrete state formulations are interested interested in and then sort of break down into if it's continuous then most of it would either be writing the the coordinates of motions themselves or they would have to use like some sort of like um um a neural network to approximate that continuous distribution of interest out or the discrete state which carl's written out um and if if they have questions about the discrete state um i'm i'm happy to take emails as well if anyone has or the the continuous state state um state space as well thanks for the distinction and um it's such a large difference between the matlab code which we got to walk through with ryan smith and christopher white which is doing matrix multiplication and such and then here comes the neural networks and it's offering sounds like new opportunities with high dimensionality and continuous variables but also a lot of new challenges so what is the essence that's shared by the matrix form and by this more machine learning style because for some people um it might be splitting a hair quite literally the difference between two computer languages when they're thinking about active inference from a ecological psychology or in an active philosophical or an embodied performance perspective all backgrounds that converge on active inference and so to somebody outside the the strand of hair what is it that we can really distill that is core active inference and i wrote down a few things so that you had said what are those core pieces that allow us to dive into the matrix mode with matlab or into the neural network mode with like python um so i think it comes

down to my summary slide the way i think about it so the active inference the core ingredients are formulating the gender model and here formulating the general model is the parameterization of the general model so either you can use the discrete stage categorical distributions or you can use a more continuous state formulation and again the neural network formulation is a specific instantiation of that so that would be one way of sort of differentiating that the second one is the optimization of the objective functions in play so in the the matlab code we're doing gradients to send using mean field message passing algorithm so specific formulation that's been introduced in a couple of papers and i specifically didn't walk through that or uh you're doing back propagation to actually calculate the or lambda distributions and then you're just solving those distributions that you have so it sort of depends on which formulation you're um like um it depends on how you optimize those objectives either you're like taking the implicit forward model or you're taking an explicit gender model um and one thing i forgot to mention is that um alex shams and connor hines they've been working on a discrete state formulation of um active inference in python which might be in of interest for people who want to like focus on one specific language that can do like the more high end or high dimensional stuff and the discrete state formulation that carl has um i i know they're also looking for people to work on the code base if anyone's interested so it's called infer activity i think um but that's on github as well if anyone's interested cool um any sort of last thoughts or comments from either of you um no i think i'm okay so what do you reckon i think we're kind of noticing a bit of a um an increase in interest in active inference like just to give you an example like maybe areas where even active inference would have been considered before like robotics um you're now beginning to see more and more of it so i think if you do want to get involved in it now is a particularly good time great call philip and i'll um just re-recommend the excellent paper that we're discussing it's in the description of this video really appreciate both of you for joining you're always welcome to come on to be speaking about a paper you've authored or not but again thanks a lot uh to nora and phillip and i hope to see you again on a future active inference stream thank you thank you for inviting us bye-bye see you daniel peace bye awesome stop the stream great conversation thanks a lot to philip and noor really